

Seat No:

Enrollment No:

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech, Summer 2025-26 Examination

Semester: 5

Subject Code: 303105301

Subject Name: AWS Fundamental

Date: 12-05-2026

Time: 10:30 AM To 01:00 PM

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Define Hybrid Cloud. b) Mention two benefits of Hybrid Cloud. c) Give one example of Hybrid Cloud usage.	06	CO1	BT1
	B. a) Define AWS Elastic Beanstalk. b) Mention one feature of Elastic Beanstalk. c) State one advantage of using Elastic Beanstalk.	06	CO3	BT2
Q.2	A. Explain the features of Amazon S3 and describe one practical use case, highlighting why S3 is suitable for it.	04	CO1	BT4
	B. Explain AWS Global Infrastructure, highlighting the difference between regions, availability zones, and edge locations, and discuss one benefit of this architecture.	05	CO1	BT1
	OR			
	B. Define VPC security and explain why it is important for deploying resources in the cloud.	05	CO2	BT1
Q.3	A. Define Serverless Computing and explain two key advantages in cloud applications, explain automatic scaling, cost efficiency in brief.	04	CO3	BT2
	B. Describe the steps to create and configure a Lambda function.	05	CO3	BT4
	OR			
	B. Sneha needs to monitor an EC2-based web application to ensure high availability and performance. Tasks: Launch an EC2 instance (Amazon Linux 2023, t2.micro) and install Apache web server. Enable CloudWatch monitoring and create an alarm for CPU utilization exceeding 70%. Configure SNS notifications for the alarm. Simulate high CPU load and verify that the alarm triggers and notifications are received.	05	CO5	BT1

SECTION - B

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Which security group rules are required to allow web access? b) Which protocol is used for web access? c) State the default port number for a web server.	06	CO2	BT2
	B. (a) What is a read replica in Amazon RDS? (b) Mention one reason for using a read replica. (c) State one difference between Multi-AZ deployment	06	CO5	BT1

	and a read replica.			
Q.2	A. Evaluate the suitability of Amazon Redshift for business intelligence workloads.	04	CO5	BT2
	B. Implement Elastic File System (EFS) with multiple EC2 instances. Tasks: • Create an EFS file system. • Launch two EC2 instances. • Mount the EFS file system on both instances. • Verify shared file accessibility.	05	CO3	BT3
	OR			
	B. Design a serverless event-driven architecture for processing user uploads. Tasks: • Create an S3 bucket user-uploads for storing files. • Create a Lambda function that processes each uploaded file (e.g., resizing images or converting formats). • Configure S3 Event Notification to trigger the Lambda function on ObjectCreated. • Store the processed output in a separate S3 bucket processed-uploads.	05	CO3	BT5
Q.3	A. Differentiate between horizontal scaling and vertical scaling.	04	CO5	BT6
	B. Explain the five essential characteristics of cloud computing as defined by NIST, giving one example for any one characteristic.	05	CO1	BT3
	OR			
	B. Explain the different types of Elastic Load Balancers (ELB) available in AWS and their primary use cases.	05	CO3	BT3